The two-step F Test

One option to replace sicegar's AIC threshold of -10 is to use F-tests. An F-test is a type of hypothesis test, often used to determine if a complex model can perform significantly better than a simple model. This section will propose two different F-tests that can be implemented sequentially to create one process. This process may change the final model decisions of sicegar.

The first F-test we will perform will compare the effectiveness of fitting a flat line to the dataset versus a sicegar model. Two separate tests will be performed, one that utilizes the sigmoidal model, and one that utilizes the double sigmoidal model. Without loss of generality, we will discuss the case for which the sigmoidal model is used. The null hypothesis for this test is that the flat line fits the data as well as the sigmoidal model does. The alternative hypothesis is that the sigmoidal model fits the data better than the flat line does. The selected alpha level is 0.01 for this F-test, which was determined empirically. The F-statistic for this test looks like this:

$$\frac{\frac{\sum(y_i - \bar{y})^2 - \sum(y_i - \hat{y}_i)^2}{p-1}}{\frac{\sum(y_i - \hat{y}_i)^2}{p}}$$

This F statistic follows an F distribution (possible reference:https://sites.duke.edu/bossbackup/files/2013/02/FTestTutorial.pdf).

If the data has a flat line of best fit, it should not be categorized as sigmoidal. In this case, the sigmoidal model will become very flat. It will have a low *slopeParam_Estimate* and a small difference between the *maximum_Estimate* and the *h0_Estimate.* This means that the SSR of the flat line and the SSR of the sigmoidal model will be similar, resulting in a small numerator in the F statistic. Overall, the F statistic will be low enough to result in a p-value greater than 0.01. If the p-value is greater than 0.01, then we will deem the dataset to be not sigmoidal.

The same process is applied for a double sigmoidal model. That is, the F-test is performed utilizing a flat line and a double sigmoidal model, and an F statistic and p-value are obtained. For p-values greater than 0.01, the dataset will be deemed not double sigmoidal.

At this point in the process, the sigmoidal and double sigmoidal classification may have both been ruled out by the first F-test. In this case, the categorization process will eventually label the dataset as ambiguous. If either (or both) of the model classifications are still in play, a second F-test will be performed on the dataset. This will be a lack of fit F-test, which will test whether each of the model classification possibilities are good fits for the dataset.

Once again, without loss of generality, we will discuss the case for which the test is performed for the sigmoidal model. The null hypothesis is that the sigmoidal model fits the data well, while the alternative hypothesis is that the sigmoidal model does not fit the data well. The F statistic is as follows:

$$\frac{\frac{[\sum r_i(\hat{y}_i - \bar{y}_i)^2]}{u-p}}{\frac{\sum(y_i - \bar{y}_i)^2}{n-u}}$$

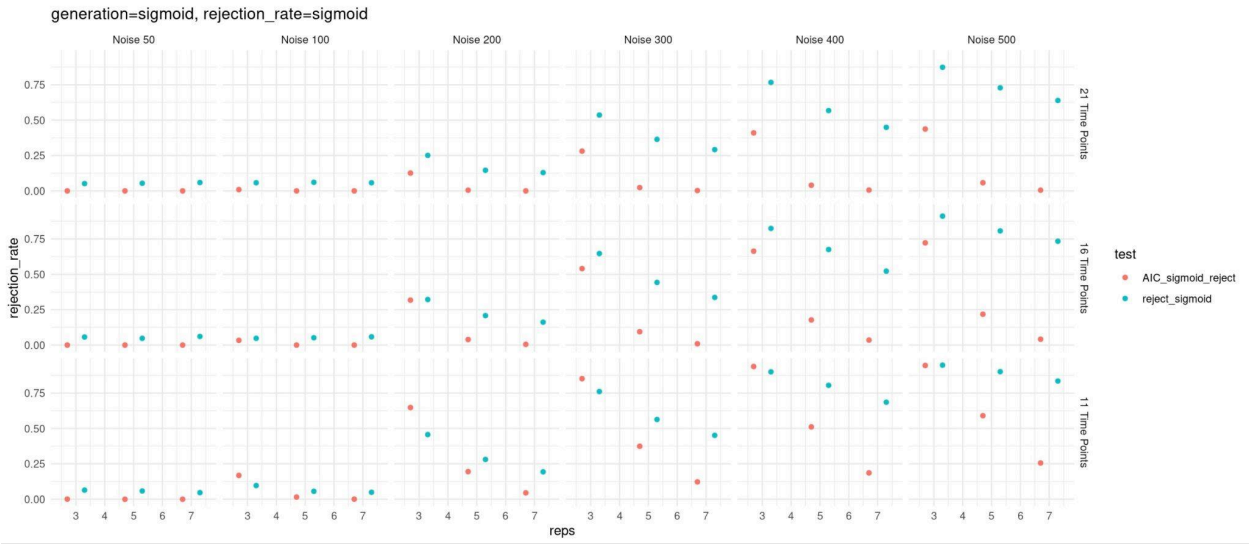
r = number of replicates
u = number of unique time points
p = number of model parameters
n = number of total observations

This F statistic follows an F distribution (possible reference: https://webpages.math.luc.edu/~tobrien/research/IWSM95.pdf).

Since time course gene expression data has replicates, we can find the average intensity value at each time point. This value can be subtracted from each observation at that time point. The resulting values are squared, summed, and then summed across time points. This quantity is the “pure error”, and the resulting value is divided by the degrees of freedom to form the denominator of the test statistic. The numerator of the statistic subtracts the mean of the replicate values at a time point from the fitted sigmoidal model value at that time point. This quantity is squared, and then multiplied by the number of replicates at that time point. The results are summed across time points, and that quantity is divided by degrees of freedom to create the numerator of the test statistic. We also perform the lack of fit F test on the double sigmoidal model. This test statistic also follows an F distribution.

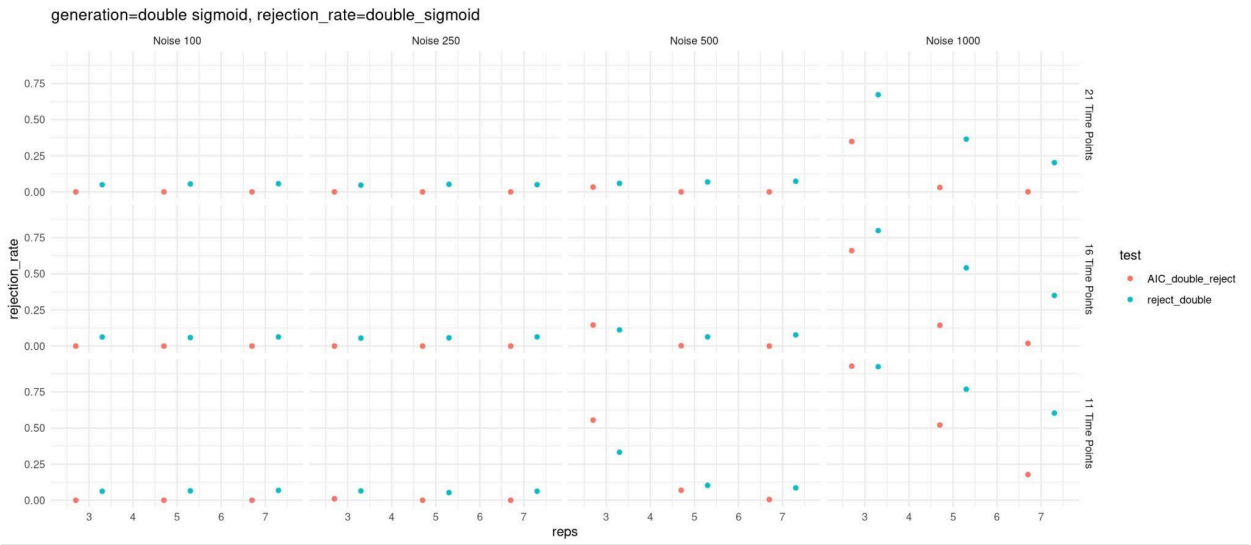
To summarize, the first F-test in the process does not permit data with a flat line of best fit to be classified as sigmoidal or double sigmoidal. When the data does not have a flat line of best fit, the second F-test determines if the sigmoidal model is a good fit for the data, and if the double sigmoidal model is a good fit for the data.

To compare the AIC-threshold of -10 to the two-step process proposed above, we ran simulations. In the simulation below, data was generated from sigmoidal models with many different parameter combinations. The h_0 and maximum values were fixed at 1500 and 1930 respectively. The number of replicates at each time point, number of time points, noise level, midpoint and slope parameter were all varied. Each blue/orange dot represents 12 types of sigmoidal models, each generated with different parameters. Every parameter combination was repeated 100 times, so each dot represents the rejection rate out of 1200 sigmoid models. The AIC value for the sigmoidal and double sigmoidal models was collected, and the proportion of AIC values above -10 was calculated. The p-values for each of the F-tests were also collected. The proportion of times that the sigmoidal model was rejected by either of the F tests was recorded. These proportions are plotted below:



The rejection rate of the two-step process is approximately 5% for low noise levels, which is expected. The AIC threshold will never reject these obvious sigmoids. However, as the noise level grows, the sigmoid model becomes unrecognizable, and the two-step process rejects at a much higher rate than the AIC threshold. This is especially true when there are 5 or 7 replicates.

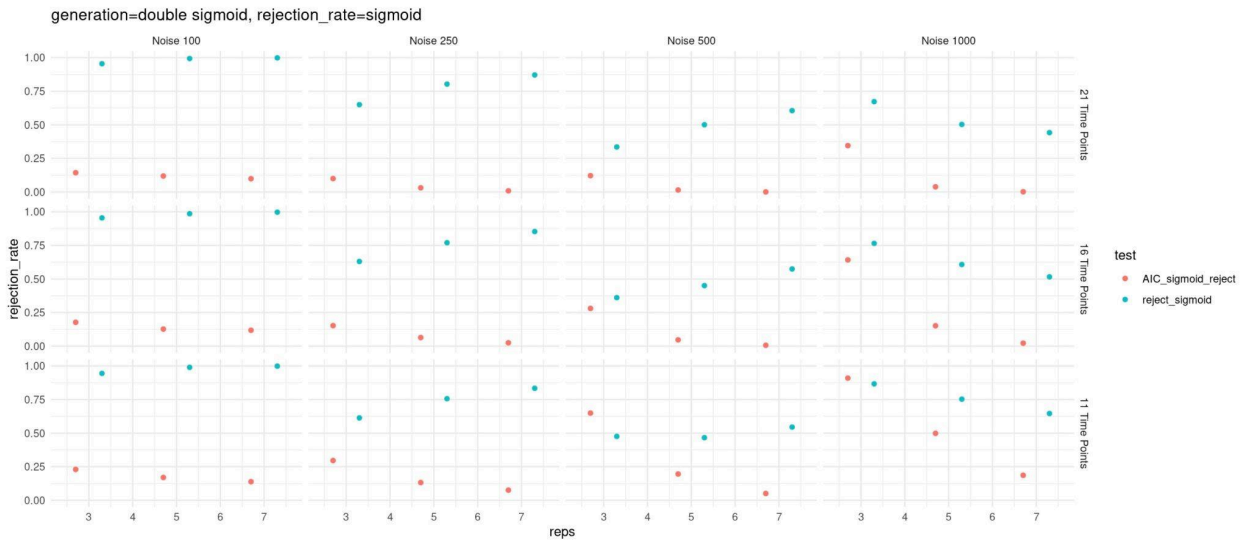
We also generated data from a double sigmoid and compared the double sigmoidal model rejection rates of the AIC threshold to the two-step F-test process. The results are below.



In this plot, the h_0 parameter was set to 700, and the maximum parameter was set to 1930. Each blue/orange dot represents 32 types of double sigmoids, each generated with different parameter combinations. Each parameter combination was repeated 100 times, so each blue/orange dot represents the average rejection rate over 3200 double sigmoidal models. The results of this plot can be interpreted similarly to the results of the previous plot. At higher noise

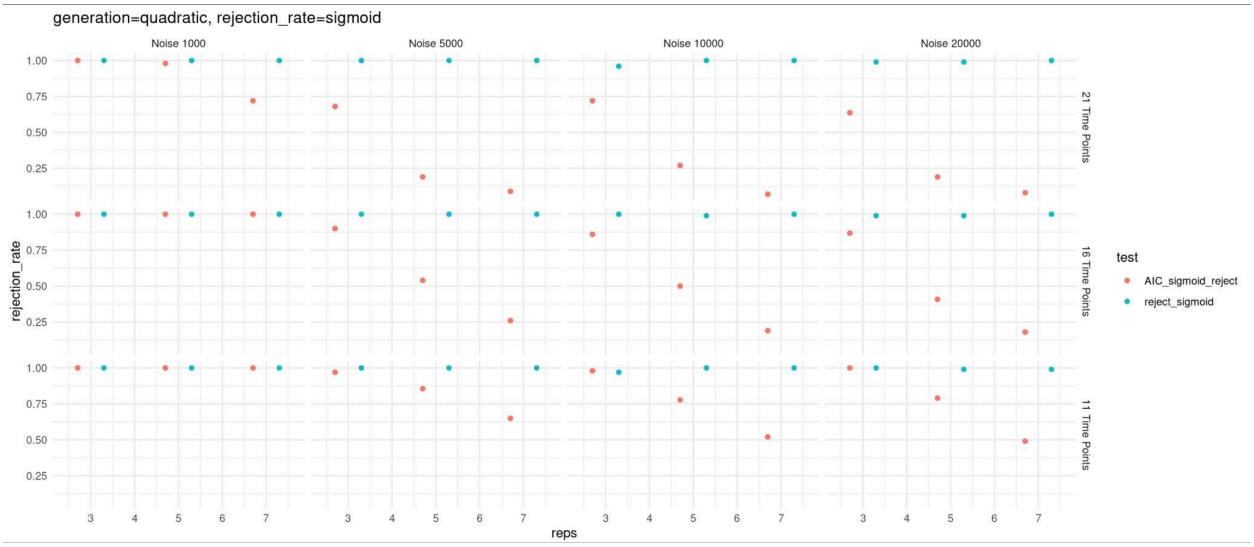
levels, when the double sigmoidal model becomes unrecognizable, the two-step F-test process rejects the double sigmoidal model at a higher rate than the AIC threshold. The AIC threshold is again less powerful at rejecting medium to high-noise double sigmoidal models when there are 5 or 7 replicates.

When generating from a double sigmoidal model, the rejection rate of the sigmoidal model should be high. This scenario is presented below.



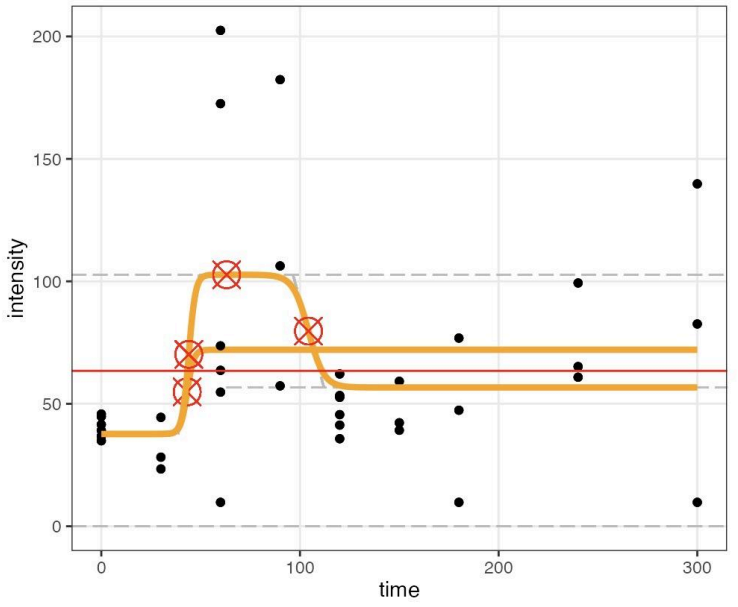
Expectedly, the two-step process has high rejection rates of the sigmoidal model for double sigmoidal data. However, the AIC threshold often fails to reject sigmoidal models in this case.

We also generated data from a quadratic, and looked at how often each model was rejected by the AIC threshold compared to the two-step F-test process. The rejection rates of the sigmoidal model for this scenario are below.

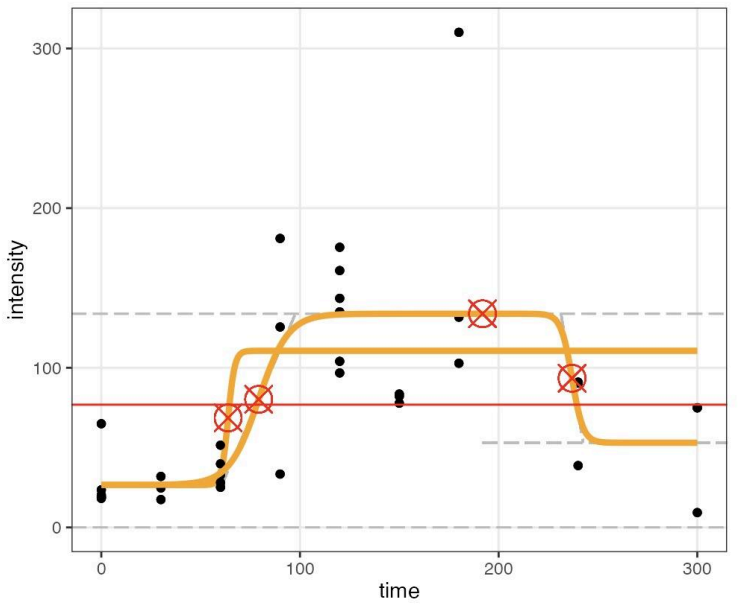


The two-step process rejects the sigmoidal model at a rate near 100% every time. The AIC threshold is sensitive to the noise level, number of replicates and number of time points. With many time points, a high noise level, and a high number of replicates, the AIC threshold rejects very few of the sigmoidal models.

These plots demonstrate that the two-step F-test works how we expect it to. (The code to run these simulations is in the “Final Simulations” folder, which is in the etcn2024 folder for the HPC). We can now demonstrate how it performs on real data. (See the “GETTING_RESULTS.qmd” file in the GitHub repository to find the code to run the tests on a single gene). Below is a dataset where the two-step F-test process rejects both models, while an AIC threshold of 5 lets them through. Specifically, both models fail the first F-test (the classic F-test).



Next is a dataset where both models pass the first F-test but fail the second F-test (the lack-of-fit F-test), while an AIC threshold of 5 lets both of them through.



Currently, we can only see how the two-step F-test process differs from the AIC threshold, but we do not fully know how this affects the final model decision. However, we can pick out some information. If the two-step F-test process rejects both models, we know the final model decision will be “ambiguous”. But the next steps will be to create a new version of fitAndCategorize that replaces the AIC threshold with the two-step F-test process. Then, we can run a dataset through both the existing fitAndCategorize function (which uses the AIC threshold) and the new fitAndCategorize function (which uses the two-step F-test process) and compare the final model decisions.